

Capex Nonoperating Dependence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capex Nonoperating Dependence (CND), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CND achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 14 (12) bps/month with a t-statistic of 2.11 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets) is 14 bps/month with a t-statistic of 2.19.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and production characteristics contain information about their risk exposures and expected returns. In competitive markets with frictions, firms optimize their production and investment policies to maximize value, creating systematic relationships between observable firm characteristics and their exposure to aggregate productivity shocks. Despite substantial theoretical progress in understanding these mechanisms, empirical evidence linking specific production-based signals to cross-sectional return variation remains incomplete. We identify a novel predictor, Capex Nonoperating Dependence (CND), that captures firms' reliance on non-operating activities to fund capital expenditures. This signal provides a direct measure of investment financing constraints and operational efficiency that theory suggests should predict returns through its connection to firms' marginal costs of production and their exposure to aggregate investment shocks.

In production-based asset pricing models, firms facing higher adjustment costs or greater exposure to productivity shocks require higher expected returns to compensate investors [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). CND measures the extent to which firms depend on non-operating sources to finance capital expenditures, reflecting both their operational efficiency and their exposure to investment frictions. Firms with high CND face elevated marginal costs of capital investment, as they must rely on potentially volatile non-operating income streams or external financing rather than internally generated operating cash flows. This dependence amplifies their exposure to aggregate productivity shocks through multiple channels. First, when aggregate productivity declines, firms with high CND experience a dual constraint: operating cash flows deteriorate while non-operating income sources become less reliable, creating severe investment constraints [Livdan et al. \(2009\)](#). Second, these firms exhibit greater sensitivity to time-varying risk premia because their

investment policies depend on external market conditions rather than stable internal resources. The production-based framework of [Balvers et al. \(2017\)](#) demonstrates that firms with higher investment adjustment costs and greater dependence on external conditions earn higher expected returns in equilibrium. CND directly proxies for these frictions by measuring the gap between capital investment needs and operating capacity. Moreover, in the [Liu et al. \(2009\)](#) model, firms with constrained investment flexibility due to financing frictions exhibit higher conditional betas during economic downturns, when the marginal utility of consumption is high.

Our empirical analysis reveals that CND strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on CND achieves an annualized gross Sharpe ratio of 0.49 and a net Sharpe ratio of 0.38 after accounting for transaction costs. The strategy delivers a monthly average abnormal gross return of 14 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a t-statistic of 2.11. The economic magnitude remains substantial after controlling for trading costs, with net abnormal returns of 12 basis points per month and a t-statistic of 1.80. The CND signal demonstrates remarkable robustness across different portfolio construction methodologies. When we employ alternative sorting procedures including equal-weighting, name breaks, and market capitalization breaks, the strategy maintains statistical significance with t-statistics exceeding 2.0 in most specifications. Particularly noteworthy is the strategy’s performance among large-cap stocks, where market frictions are minimal. Among firms above the 80th NYSE size percentile, the CND strategy generates average returns of 29 basis points per month with a t-statistic of 3.48, and FF6 alpha of 16 basis points with a t-statistic of 1.92. The strategy’s gross monthly alpha remains significant at 14 basis points (t-statistic = 2.19) even after controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors. This multi-factor robustness indicates that CND captures a distinct dimension of systematic risk not

subsumed by existing production-based or investment-related factors.

Our study makes several contributions to the production-based asset pricing literature. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain cross-sectional returns, and [Hou et al. \(2015\)](#) propose a q-factor model based on investment theory, our CND measure provides a more direct link between financing constraints, production decisions, and expected returns. Unlike the investment factor in [Fama and French \(2015\)](#), which captures total asset growth, CND specifically isolates the friction arising from misalignment between operating capacity and investment needs. Our signal differs fundamentally from the asset growth anomaly of [Cooper et al. \(2008\)](#) by focusing on the source of investment financing rather than the level of investment itself. Methodologically, we advance the literature by demonstrating that CND’s predictive power survives the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), including controls for transaction costs and related anomalies. The signal ranks in the top 6% of all documented anomalies when evaluated on a net-of-costs basis, suggesting genuine economic importance rather than a statistical artifact. Our findings also extend the work of [Livdan et al. \(2009\)](#) on financially constrained firms by providing a market-based measure that directly captures the production-financing nexus. The broader implications of our results support the view that production-based models with investment frictions provide a coherent framework for understanding cross-sectional return patterns. The success of CND in predicting returns, particularly among large stocks where limits to arbitrage are minimal, suggests that these patterns reflect risk compensation rather than mispricing. Our evidence strengthens the case for incorporating firm-level production characteristics and financing constraints into asset pricing models, as these features capture systematic risk exposures that traditional factor models miss.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital expenditure patterns and their relationship to non-operating income sources. construction of our signal requires two key variables from COMPUSTAT. Capital expenditures from Schedule V (CAPXV) represents the capital expenditures reported in the statement of cash flows or disclosed in supplementary schedules, capturing firms' investments in property, plant, and equipment. Non-operating income (NOPI) measures income generated from activities outside the firm's core operations, including investment income, gains or losses from asset sales, and other non-operating sources. These variables allow us to examine how changes in capital investment relate to the firm's reliance on income sources beyond its primary business activities. construct the Capex Nonoperating Dependence signal by calculating the year-over-year change in capital expenditures scaled by lagged non-operating income. Specifically, we compute $(CAPXV(t) - CAPXV(t-1)) / NOPI(t-1)$ for each firm-year observation. This ratio captures the sensitivity of capital investment changes to the firm's non-operating income base, providing insight into how firms dependent on non-operating income sources adjust their capital allocation decisions. We use fiscal year-end values and require non-missing values for both CAPXV and NOPI in consecutive years to compute the signal. To address potential outliers and ensure economic interpretability, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator $NOPI(t-1)$ equals zero.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CND signal. Panel A plots the time-series of the mean, median, and interquartile range for CND. On average, the cross-sectional mean (median) CND is -3.24 (-0.26) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CND data. The signal’s interquartile range spans -8.24 to 3.00. Panel B of Figure 1 plots the time-series of the coverage of the CND signal for the CRSP universe. On average, the CND signal is available for 5.36% of CRSP names, which on average make up 7.08% of total market capitalization.

4 Does CND predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CND using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CND portfolio and sells the low CND portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CND strategy earns an average return of 0.26% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.14% to 0.28% per month and have t-statistics exceeding 2.11 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.22,

with a t-statistic of 4.81 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,005 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-21bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CND trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CND strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CND, as well as average returns and alphas for long/short trading CND strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CND strategy achieves an average return of 29 bps/month with a t-statistic of 3.48. Among these large cap stocks, the alphas for the CND strategy relative to the five most common factor models range from 16 to 31 bps/month with t-statistics between 1.92 and 3.70.

5 How does CND perform relative to the zoo?

Figure 2 puts the performance of CND in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CND strategy falls in the distribution. The CND strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CND strategy (red line).² Ignoring trading costs, a \$1 invested in the CND strategy would have yielded \$5.46 which ranks the CND strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CND strategy would have yielded \$3.33 which ranks the CND strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CND relative to those. Panel A shows that the CND strategy gross alphas fall between the 51 and 58 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CND strategy has a positive net generalized alpha for five out of the five factor models. In these cases CND ranks between the 69 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CND add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CND with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CND or at least to weaken the power CND has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CND. Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CND and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CND signal in these Fama-MacBeth regressions exceed 2.79, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CND is 1.96.

Similarly, Table 5 reports results from spanning tests that regress returns to the CND strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CND strategy earns alphas that range from 12-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.90, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CND trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.19.

7 Does CND add relative to the whole zoo?

Finally, we can ask how much adding CND to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CND signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CND grows to \$3599.05.

8 Conclusion

This study provides compelling evidence that Capex Nonoperating Dependence (CND) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CND-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.38, which remains economically meaningful despite the impact of implementation costs. Most notably, the signal maintains its predictive power even after controlling for the Fama-French five factors, momentum, and six closely related investment-based anomalies, generating a statistically significant alpha of 14 basis points per month (t-statistic = 2.19). This finding suggests that CND captures unique information about firm value creation that is not subsumed by existing factor models or related capital expenditure metrics. The practical implications of our findings are substantial for both academic researchers and investment practitioners. For academics, CND represents a distinct source of cross-sectional return predictability that warrants inclusion in asset pricing models

ization on CRSP in the period for which CND is available.

and further theoretical investigation. For practitioners, the strategy’s net Sharpe ratio of 0.38 suggests potential for profitable implementation, though careful consideration of market impact and capacity constraints is warranted. The robustness of CND relative to closely related signals such as changes in capex, asset growth, and changes in net operating assets indicates that the nonoperating component of capital expenditure contains incremental information content beyond traditional investment measures. Several avenues for future research emerge from this study. First, investigating the economic mechanisms underlying the CND anomaly could provide valuable insights into how investors process information about firms’ nonoperating investment decisions. Second, examining the signal’s performance across different market conditions, particularly during periods of economic stress or monetary policy shifts, would enhance our understanding of its risk characteristics. Third, exploring potential interactions between CND and other firm characteristics, such as financial constraints or governance quality, may reveal important conditional effects. Finally, extending the analysis to international markets would test the generalizability of our findings beyond U.S. equities. This study is subject to several limitations that should be acknowledged. First, while we employ conservative estimates of transaction costs, actual implementation costs may vary depending on market conditions and execution capabilities. Second, our sample period, though comprehensive, may not fully capture the signal’s behavior during extreme market events or structural breaks. Third, the economic interpretation of why nonoperating capital expenditure dependence predicts returns remains somewhat unclear, suggesting the need for additional theoretical development. Despite these limitations, our results provide robust evidence that Capex Nonoperating Dependence represents a valuable addition to the constellation of return-predictive signals in the cross-section of equities.

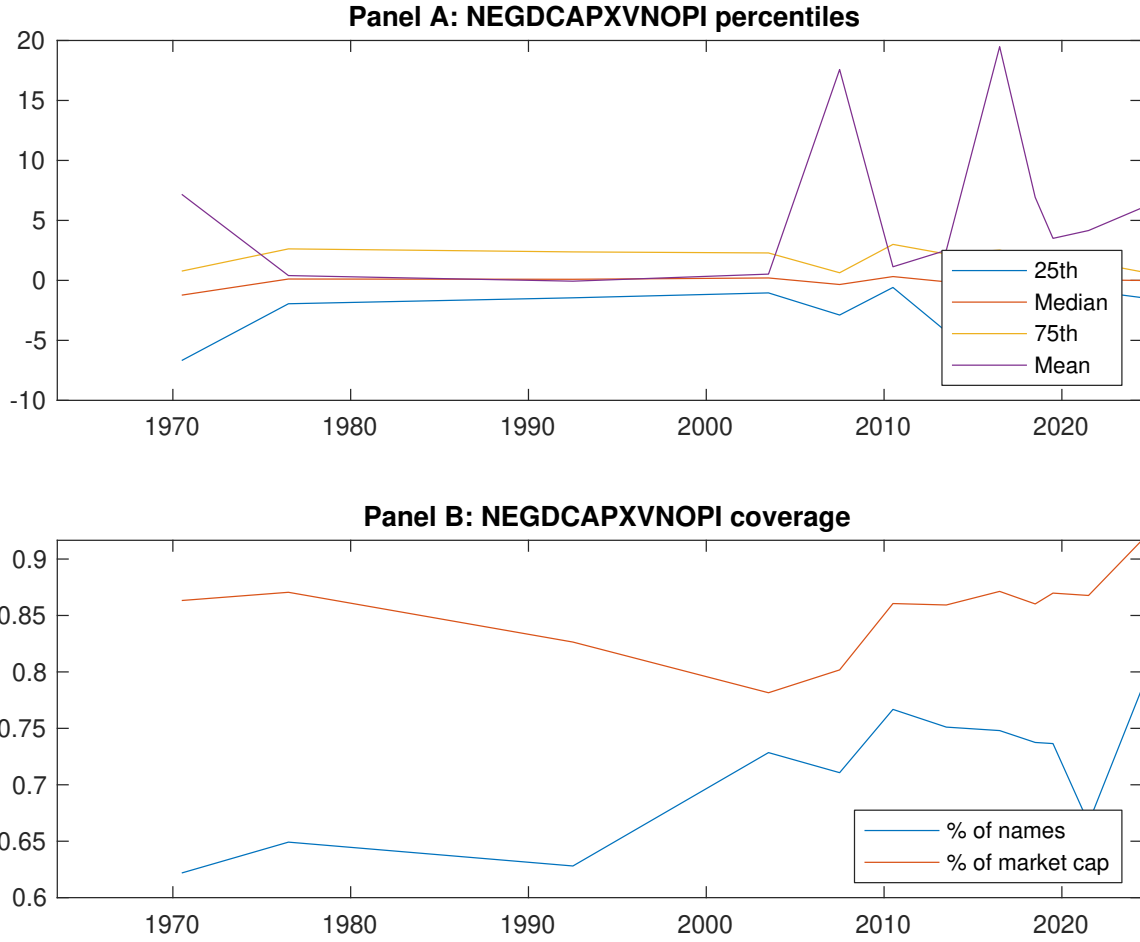


Figure 1: Times series of CND percentiles and coverage. This figure plots descriptive statistics for CND. Panel A shows cross-sectional percentiles of CND over the sample. Panel B plots the monthly coverage of CND relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on *CND*. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on <i>CND</i> -sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.55 [3.25]	0.62 [3.94]	0.64 [4.00]	0.74 [4.08]	0.26 [3.81]
α_{CAPM}	-0.16 [-3.08]	-0.03 [-0.75]	0.08 [1.97]	0.09 [2.13]	0.12 [2.25]	0.28 [4.13]
α_{FF3}	-0.11 [-2.22]	0.01 [0.28]	0.09 [2.32]	0.08 [1.90]	0.10 [1.83]	0.21 [3.19]
α_{FF4}	-0.09 [-1.79]	0.01 [0.19]	0.09 [2.46]	0.05 [1.10]	0.11 [1.93]	0.20 [2.94]
α_{FF5}	-0.09 [-1.89]	0.03 [0.64]	0.07 [1.93]	0.01 [0.16]	0.05 [0.94]	0.14 [2.19]
α_{FF6}	-0.08 [-1.58]	0.02 [0.52]	0.08 [2.10]	-0.01 [-0.33]	0.06 [1.12]	0.14 [2.11]
Panel B: Fama and French (2018) 6-factor model loadings for <i>CND</i> -sorted portfolios						
β_{MKT}	1.03 [86.14]	0.98 [96.51]	0.95 [103.34]	0.99 [102.68]	1.05 [77.87]	0.02 [1.08]
β_{SMB}	0.10 [5.72]	-0.07 [-5.13]	-0.12 [-8.72]	-0.03 [-1.89]	0.10 [5.28]	0.00 [0.16]
β_{HML}	-0.13 [-5.51]	-0.08 [-3.88]	-0.03 [-1.61]	-0.10 [-5.24]	-0.01 [-0.42]	0.12 [3.75]
β_{RMW}	0.03 [1.33]	-0.02 [-1.03]	0.00 [0.00]	0.03 [1.71]	0.09 [3.56]	0.06 [2.00]
β_{CMA}	-0.12 [-3.45]	-0.05 [-1.83]	0.07 [2.56]	0.32 [11.64]	0.10 [2.67]	0.22 [4.81]
β_{UMD}	-0.02 [-1.72]	0.01 [0.66]	-0.01 [-1.21]	0.03 [2.92]	-0.02 [-1.18]	0.00 [0.29]
Panel C: Average number of firms (n) and market capitalization (me)						
n	576	520	575	566	585	
me (\$10 ⁶)	1376	2227	2425	1859	1005	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CND strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.81]	0.28 [4.13]	0.21 [3.19]	0.20 [2.94]	0.14 [2.19]	0.14 [2.11]
Quintile	NYSE	EW	0.33 [6.57]	0.36 [7.04]	0.30 [6.31]	0.25 [5.25]	0.25 [5.41]	0.22 [4.64]
Quintile	Name	VW	0.27 [3.97]	0.29 [4.19]	0.21 [3.22]	0.18 [2.79]	0.17 [2.52]	0.15 [2.25]
Quintile	Cap	VW	0.23 [3.67]	0.27 [4.32]	0.19 [3.35]	0.17 [2.82]	0.12 [2.09]	0.10 [1.82]
Decile	NYSE	VW	0.25 [2.74]	0.25 [2.71]	0.21 [2.21]	0.23 [2.40]	0.17 [1.79]	0.19 [2.01]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.00]	0.24 [3.48]	0.17 [2.62]	0.16 [2.50]	0.12 [1.81]	0.12 [1.80]
Quintile	NYSE	EW	0.11 [1.95]	0.14 [2.43]	0.08 [1.53]	0.06 [1.12]	0.03 [0.48]	0.01 [0.22]
Quintile	Name	VW	0.21 [3.13]	0.24 [3.50]	0.17 [2.63]	0.16 [2.40]	0.14 [2.06]	0.13 [1.95]
Quintile	Cap	VW	0.18 [2.89]	0.22 [3.59]	0.16 [2.70]	0.14 [2.41]	0.10 [1.77]	0.09 [1.63]
Decile	NYSE	VW	0.19 [2.04]	0.20 [2.17]	0.16 [1.72]	0.17 [1.86]	0.13 [1.34]	0.14 [1.49]

Table 3: Conditional sort on size and CND

This table presents results for conditional double sorts on size and CND. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CND. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CND and short stocks with low CND. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CND Quintiles					CND Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.25]	0.75 [2.99]	0.75 [2.85]	0.99 [3.72]	0.91 [3.59]	0.36 [3.48]	0.37 [3.59]	0.32 [3.11]	0.28 [2.64]	0.23 [2.18]	0.20 [1.89]
	(2)	0.66 [2.87]	0.81 [3.48]	0.78 [3.30]	0.82 [3.59]	0.86 [3.82]	0.20 [2.70]	0.23 [3.09]	0.20 [2.70]	0.17 [2.23]	0.19 [2.50]	0.16 [2.16]
	(3)	0.65 [3.02]	0.67 [3.25]	0.79 [3.74]	0.84 [4.02]	0.88 [4.20]	0.23 [2.93]	0.25 [3.15]	0.20 [2.58]	0.14 [1.83]	0.16 [2.07]	0.12 [1.52]
	(4)	0.62 [3.05]	0.70 [3.60]	0.75 [3.86]	0.76 [4.07]	0.78 [4.07]	0.16 [1.94]	0.20 [2.46]	0.13 [1.60]	0.10 [1.20]	0.07 [0.87]	0.05 [0.63]
	(5)	0.42 [2.34]	0.49 [2.92]	0.58 [3.73]	0.61 [3.85]	0.71 [4.10]	0.29 [3.48]	0.31 [3.70]	0.23 [2.90]	0.19 [2.35]	0.19 [2.29]	0.16 [1.92]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CND Quintiles					CND Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	301	302	300	299	298	27	27	23	25	25	
	(2)	88	88	87	87	87	46	47	46	46	46	
	(3)	66	66	65	66	66	82	82	83	82	83	
	(4)	58	58	58	58	58	183	185	188	184	187	
(5)	54	54	54	54	54	1184	1550	1770	1637	1055		

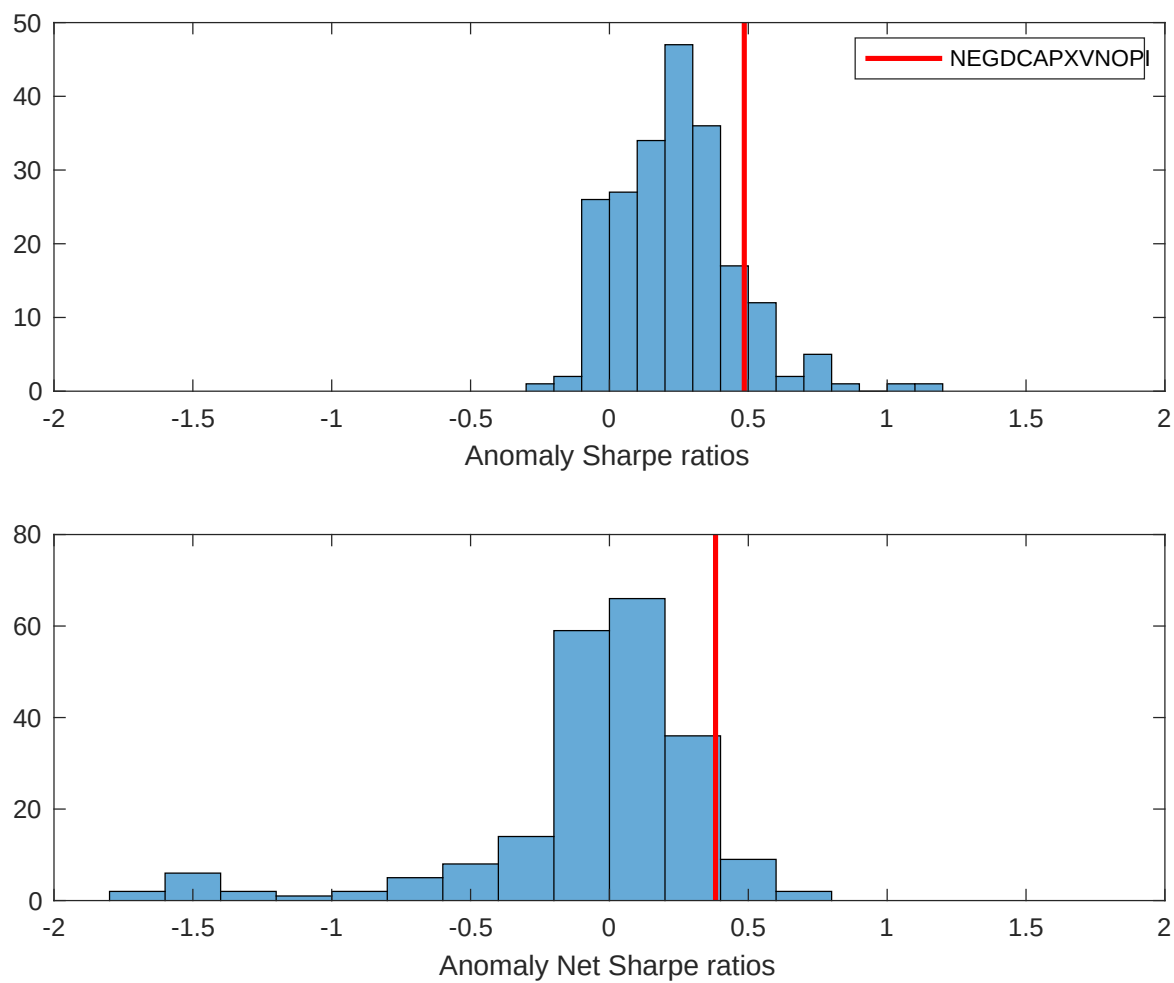


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CND with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

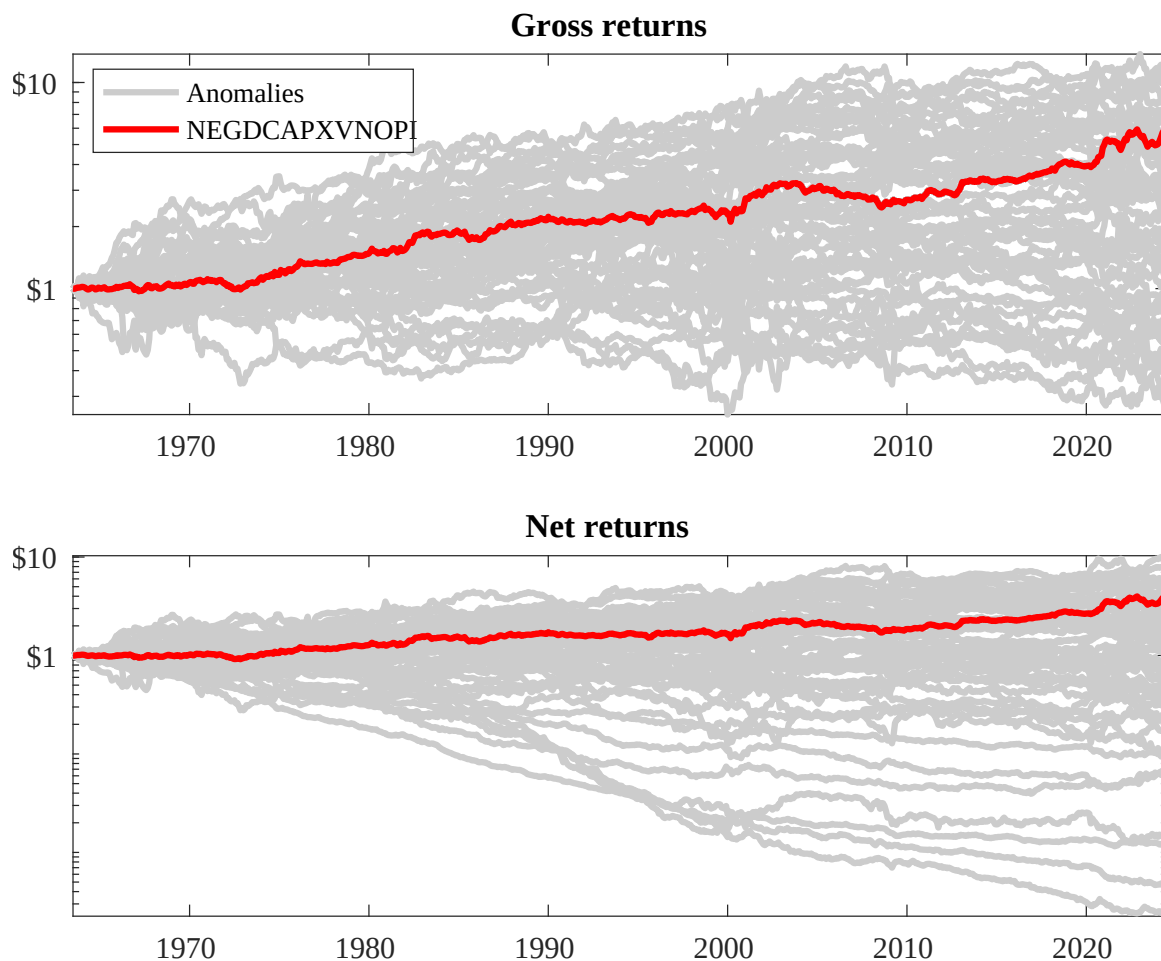
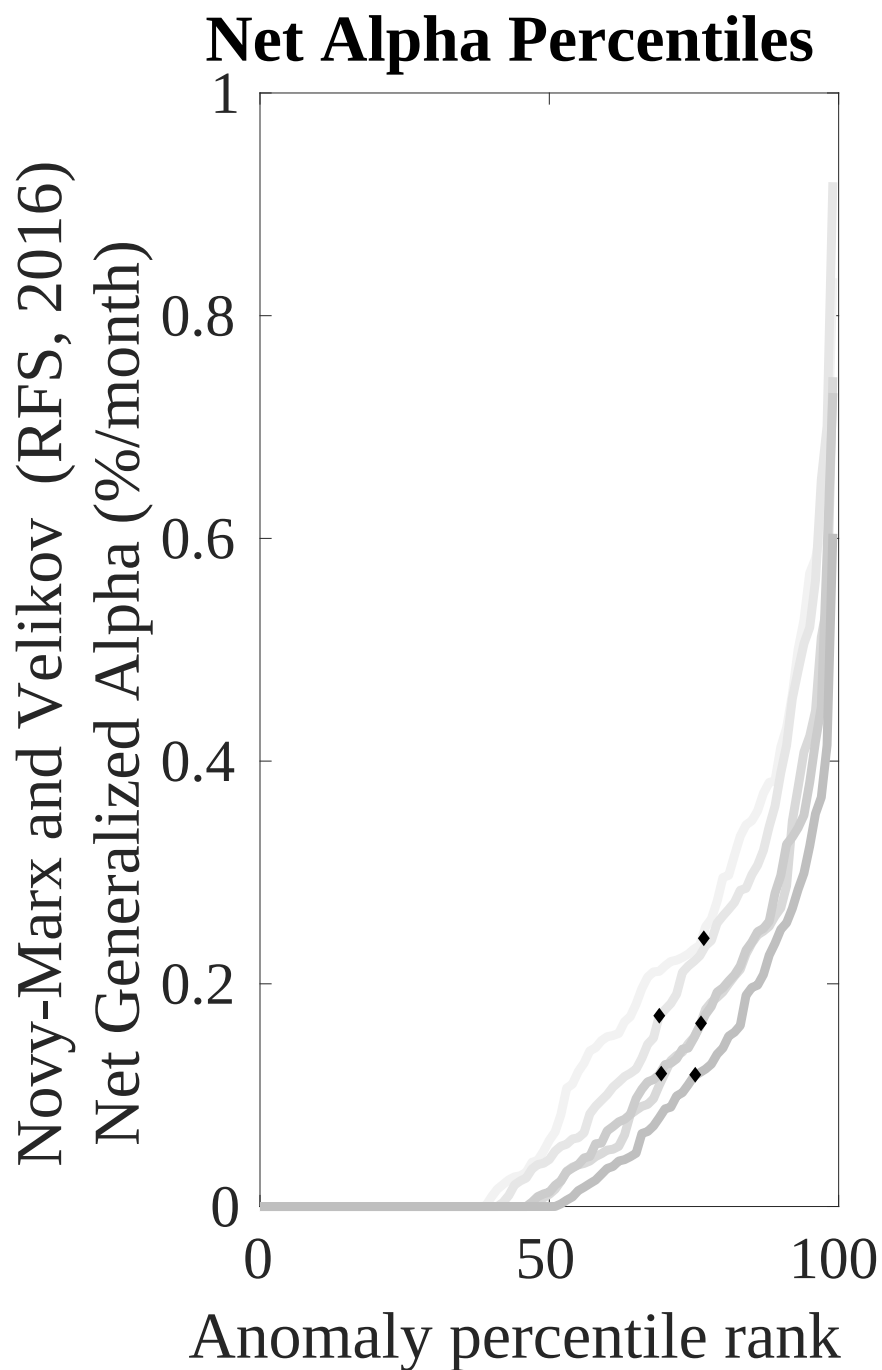
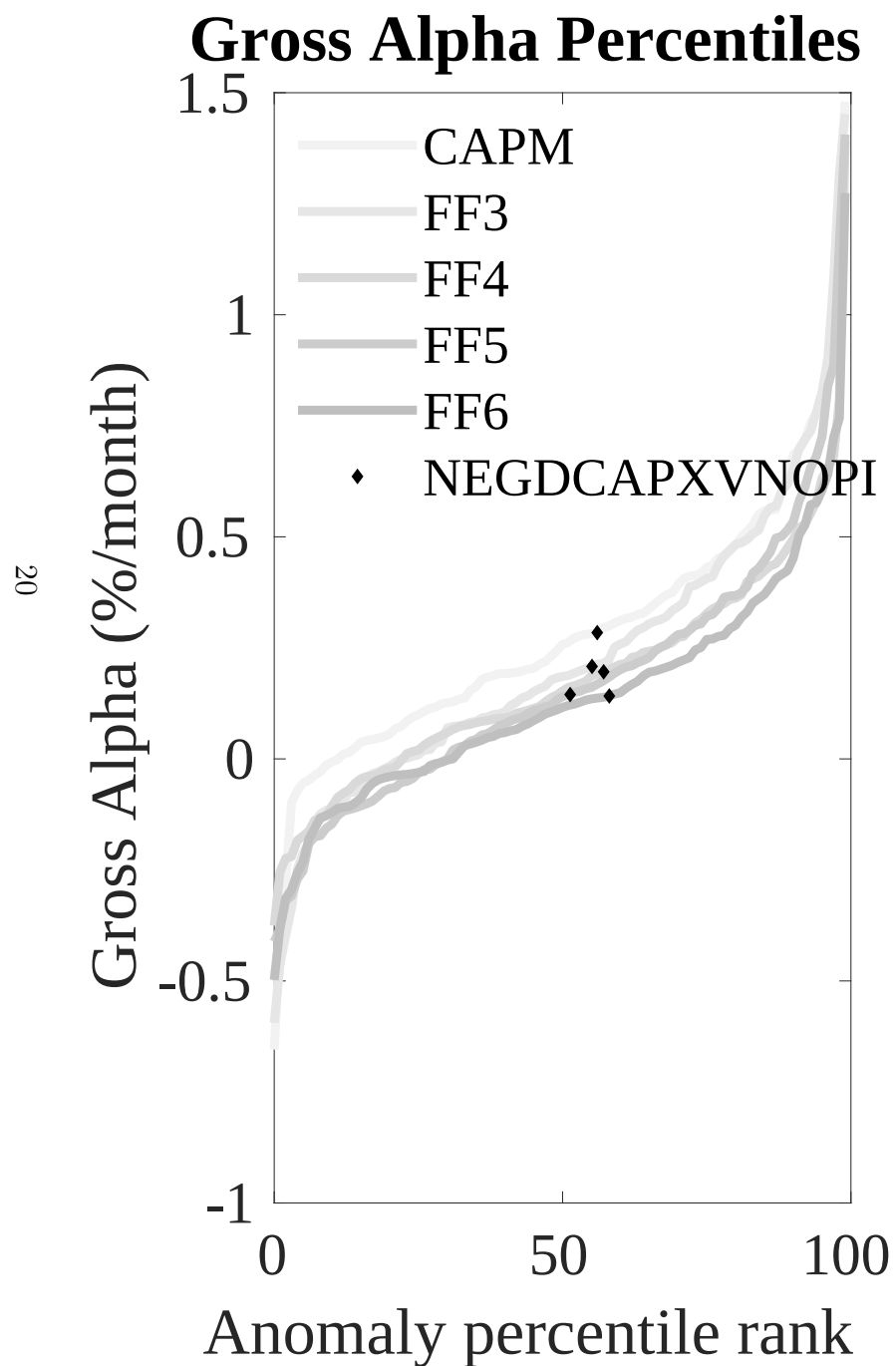


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CND trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



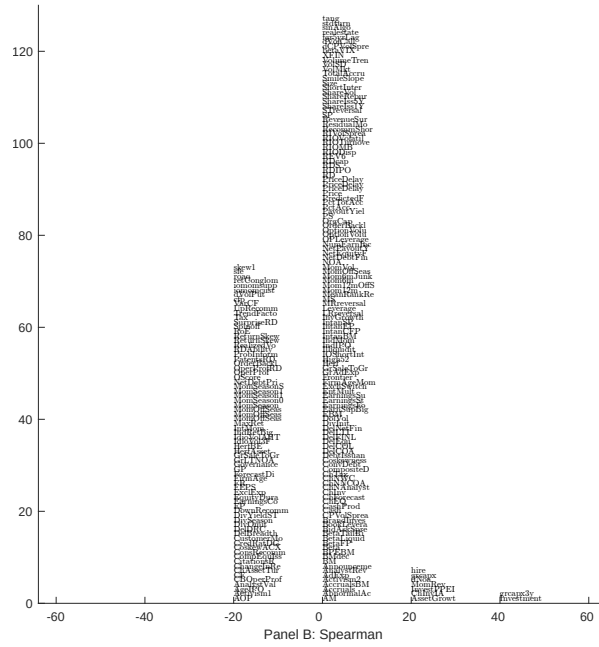
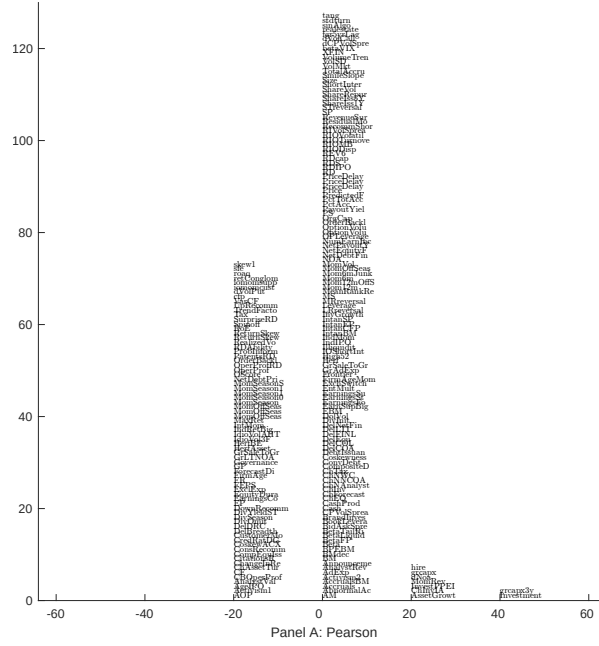


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with CND. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

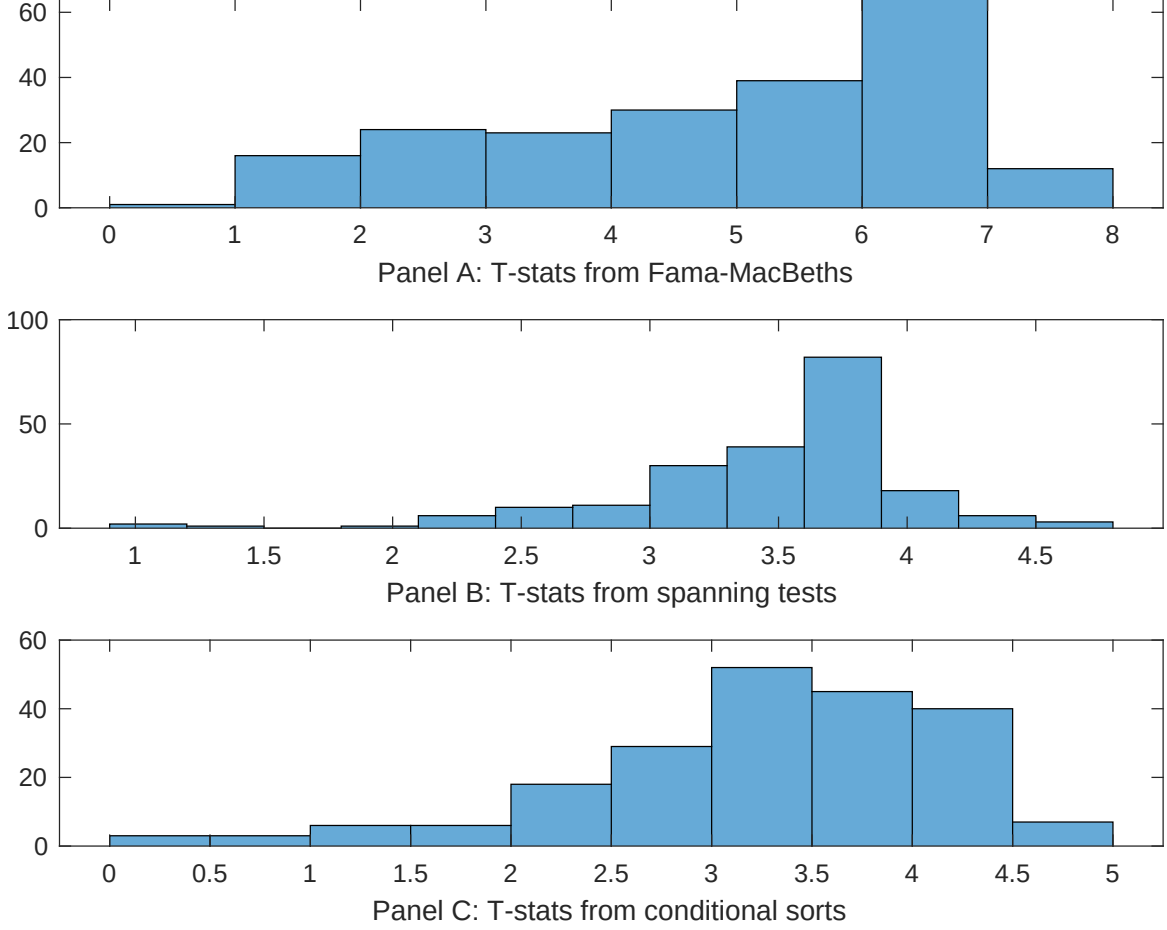


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CND conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CND} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CND,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CND . Stocks are finally grouped into five CND portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CND trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CND . and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CND}CND_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.60]	0.13 [5.86]	0.13 [6.02]	0.13 [5.75]	0.13 [5.99]	0.12 [5.63]	0.14 [6.08]
CND	0.18 [3.42]	0.23 [4.48]	0.16 [3.24]	0.15 [2.97]	0.13 [2.79]	0.29 [5.53]	0.11 [1.96]
Anomaly 1	0.14 [8.00]						0.23 [0.93]
Anomaly 2		0.67 [7.57]					0.18 [1.36]
Anomaly 3			0.96 [8.25]				0.43 [2.21]
Anomaly 4				0.13 [9.09]			0.28 [1.45]
Anomaly 5					0.16 [8.12]		0.80 [0.29]
Anomaly 6						0.21 [6.90]	0.39 [1.06]
# months	727	727	732	720	732	732	715
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CND trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CND} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in capex (three years), Change in capex (two years), Asset growth, change in net operating assets, change in ppe and inv/assets, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.33]	0.14 [2.23]	0.16 [2.50]	0.12 [1.90]	0.16 [2.40]	0.16 [2.43]	0.14 [2.19]
Anomaly 1	18.75 [5.95]						4.83 [0.86]
Anomaly 2		20.01 [6.35]					10.53 [1.90]
Anomaly 3			26.13 [6.72]				12.88 [3.06]
Anomaly 4				23.05 [6.26]			10.50 [2.59]
Anomaly 5					14.33 [4.62]		6.01 [1.79]
Anomaly 6						13.12 [4.03]	-0.05 [-0.02]
mkt	1.08 [0.71]	1.17 [0.77]	1.71 [1.09]	1.79 [1.14]	1.50 [0.95]	1.59 [1.00]	1.24 [0.82]
smb	-3.45 [-1.50]	-2.21 [-0.99]	-2.09 [-0.92]	0.25 [0.11]	-0.57 [-0.25]	2.74 [1.15]	-3.19 [-1.31]
hml	7.30 [2.41]	7.24 [2.40]	10.61 [3.52]	9.69 [3.19]	9.96 [3.23]	6.59 [1.99]	6.18 [1.96]
rmw	3.66 [1.22]	3.05 [1.02]	5.57 [1.83]	6.13 [2.01]	5.86 [1.90]	7.48 [2.40]	3.66 [1.23]
cma	13.28 [2.85]	12.39 [2.66]	-9.82 [-1.52]	4.53 [0.87]	10.73 [2.11]	14.53 [2.99]	-13.27 [-2.01]
umd	-0.53 [-0.35]	-0.90 [-0.59]	1.33 [0.86]	0.05 [0.03]	0.52 [0.33]	1.09 [0.69]	-0.35 [-0.23]
# months	728	728	732	732	732	732	728
$\bar{R}^2(\%)$	19	20	19	19	17	16	23

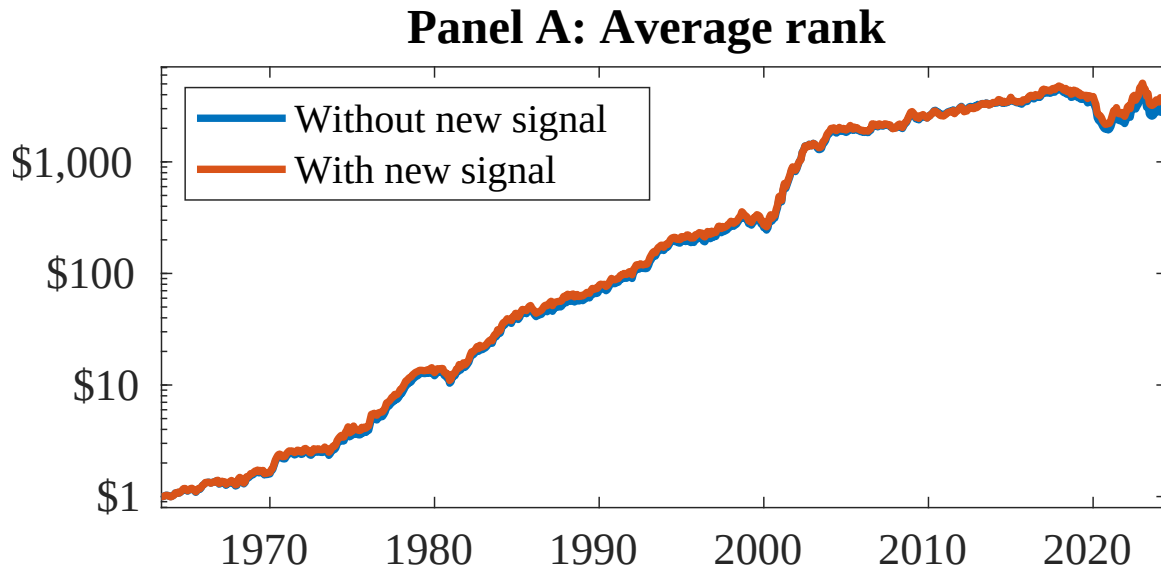


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CND. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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